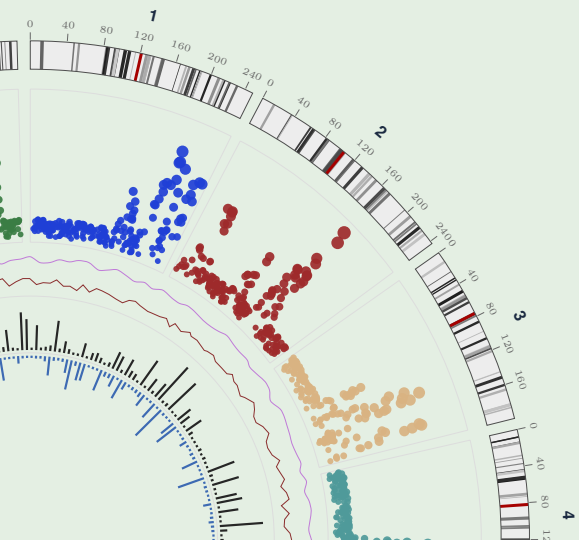
Distinguishing true signals from noise in large, high-dimensional omics datasets is key to uncovering the genetic architecture of complex traits and the molecular drivers of disease. This thesis develops scalable Bayesian inference frameworks based on Vector Approximate Message Passing (VAMP).

gVAMP jointly models complex traits across millions of genetic variants. Applied to 17 million whole-genome sequence variants from the UK Biobank, it achieves a prediction accuracy of approximately 46% for human height, the highest reported for this trait to date.

vampW brings this framework to survival analysis. In the UK Biobank Pharma Proteomics Project, it identifies 219 protein associations across 24 disease outcomes, most of which are not among the top marginal discoveries, and delivers state-of-the-art prediction of disease onset times.

Together, these methods provide novel tools for dissecting complex traits and advancing personalized risk prediction.



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